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OSSP:

Commands:

uname: Displays operating system information.

lscpu: Shows CPU details such as cores and architecture.

```
[b.deepak@BDeepaks-MacBook-Air ~ % gcc os.c -o os
[b.deepak@BDeepaks-MacBook-Air ~ % ./os
Enter a Linux command: uname

Parent Process
Parent PID: 33911
Waiting for child process...

Child Process
Child PID: 33912
Darwin
Child process completed.
b.deepak@BDeepaks-MacBook-Air ~ %
```

ps: Displays the currently running processes.

```
[b.deepak@BDeepaks-MacBook-Air ~ % ./os
Enter a Linux command: ps

Parent Process
Parent PID: 33932
Waiting for child process...

Child Process
Child PID: 33933
  PID TTY           TIME CMD
33890 ttys000    0:00.02 -zsh
33932 ttys000    0:00.01 ./os
Child process completed.
b.deepak@BDeepaks-MacBook-Air ~ %
```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system controls how the processor is used by different applications. It schedules tasks so that every running program gets CPU time efficiently.

Example:
If a web browser and a video player are running at the same time, the OS switches the CPU between them, making both applications appear to run simultaneously.

2. Memory (RAM) -> Memory Management

The operating system allocates RAM to programs based on their requirements. It also protects each program's memory to prevent interference from other applications.

Example:
When Microsoft Word and Google Chrome are open together, each application receives its own memory space without affecting the other.

3. Storage (SSD) → File System Management

The operating system manages how files and folders are stored, accessed, modified, and organized on storage devices.

Example:
When you create and save a PowerPoint presentation, the OS stores it in the selected folder and retrieves it whenever you open it again.

4. I/O Devices → Device Management

The operating system communicates with hardware devices through device drivers. It manages the exchange of data between the computer and connected peripherals.

Example:
When you connect a USB keyboard or print a document using a printer, the OS uses the appropriate driver to communicate with the device and perform the requested operation.